F-8

BABY DON'T GOT BACK

A CHAIR SELF-AWARE OF ITS LACK OF BACK REST

APSC 100 Team F-8

Re-engineering and redefining a chair for evacuation sites that can be easily manufactured, transported, setup, and deployed within limited timeframes

F-8

Assessment of Stakeholders' Needs

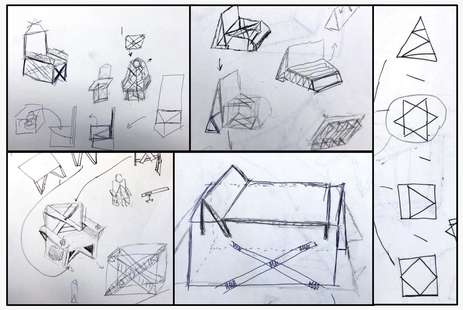
Before brainstorming early design concepts, the stakeholders' needs were taken into consideration as well as their relevance to the design. The following 5 primary stakeholders, their importance, and needs were identified

1. **Children**
   - The children will be the main users of the chair and their needs must be considered extensively
   - **Threshold needs:** Capable of holding the child safely
   - **Expressed needs:** Comfortable, interesting, portable, customizable
2. **Parents**
   - The parents will set up the chairs and care for the safety of their children
   - **Threshold needs:** Safe for child
   - **Expressed needs:** Easy set up, entertaining
3. **Non- profit organization (funders of the project)**
   - The non-profit will fund and produce the chairs, and select the final design of the chair
   - **Expressed needs:** Built from 4'x4' cardboard; 100 pieces could be packaged and shipped as a bundle, ~40 cm high; supports >60 lbs; fit for 5-8 year old children; cheap; easy and quick to distribute, build, and disassemble; long-lasting
4. **Safety Regulators**
   - The safety regulators will determine the safety, reliability, usability, and potential hazards of the chair
   - **Threshold needs:** Safe and easy to use; fits production standards; chair comes with clear instructions and labelling that will not cause misuse or health and safety concerns
   - **Expressed needs:** Easily disposable and recyclable
5. **The Evacuation Centre**
   - The evacuation centre's regulations and functions must not be affected by the chairs. The chairs should not inconvenience the evacuation centre.
   - **Expressed needs:** Compact, easy to store and distribute

Early Concepts

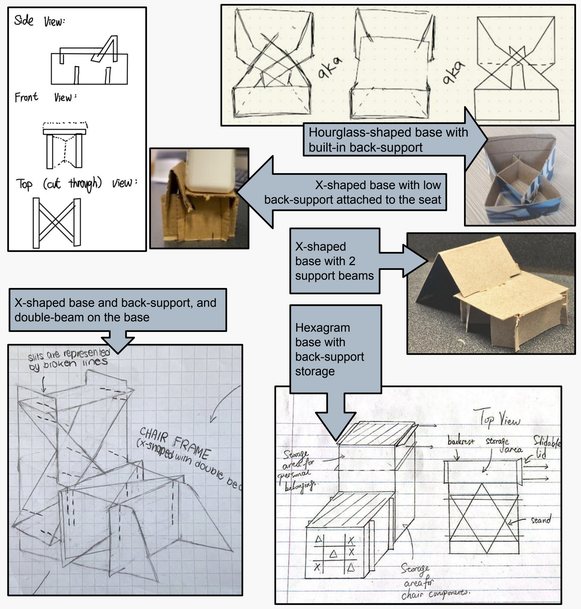
Based on the stakeholder needs as shown above, two rounds of c-sketches, as shown in Figure 1, were conducted to generate a variety of focused virtual prototypes to provide insights into the final design. Key features of the c-sketches were **low back-support**, **interlocking base** design, and **compact design**.

Figure 1. C-Sketch Prototype



Early Concepts (continued)

Figure 2. Individual Prototypes



Each group member then sketched and built an individual prototype to test various chair shapes, bases, and backrests. It was concluded that an **x-shaped** or **interlocked triangles** base were the most promising in terms of support. In addition, a **low, angled back-support** could be attached to the chair for stability.

Key Specifications

Through team meetings, the following key specifications were examined, and then goals were set for each. These were later used as criterias for the ranking and scoring process.

1. **Ease of Assembly**
   - Able to be assembled by one person
2. **Weight Supported**
   - The more the better; preferably be able to support 230 lbs
3. **Material Required**
   - The smaller the better; preferably be within 1400 in^2
4. **Comfort**
   - Comfortable; preferably scoring ~5, but back rest length is secondary
5. **Engagement Features & Aesthetics**
   - Overall design should look clean; capable of being decorated, able to score 6 to 8 points on evaluation
6. **Deviation from Ideal Seat Height and Depth Range**
   - Total deviation between 6 inches to 11 inches
7. **Seat Width**
   - The wider the better; have seat width at least 12 inches

Ranking

Two of the designs that were ranked lower during the ranking process were the **back-support storage** design and the **grid-shaped base** due to **higher cardboard consumption**. In addition, the design with **a triangle embedded in another triangle** was ranked lower due to **more parts to assemble** and **higher cardboard consumption**. Although the designs performed well in **weight support** and **additional features**, it was decided that other key design specifications needed to prioritized. Therefore, the ideas were eliminated from the scoring process.

Scoring

The scoring process was completed with a spreadsheet. Each team member scored a design using the criterias listed below in Table 1 on a scale of 1 to 10. The scores were then added and averaged for comparison. The final design chosen was the one with the highest score, a combination of two chairs, creating an hourglass-shaped base with folding back-support model.

Table 1. Scoring Results

Hourglass-shaped Base with Built-in Back-support						
Ease to Build	Weight Support	Materials	Comfort	Aesthetics	Dimensions	OVERALL AVERAGE
8.6	7.0	8.7	5.9	7.6	6.9	**7.4**

X-shaped Base with 2 Support Beams and Folding Back-support						
Ease to Build	Weight Support	Materials	Comfort	Aesthetics	Dimensions	OVERALL AVERAGE
7.9	7.0	8.0	7.6	7.4	7.1	**7.5**

X-shaped Base with Low Back-support Attached to the Seat						
Ease to Build	Weight Support	Materials	Comfort	Aesthetics	Dimensions	OVERALL AVERAGE
6.9	8.3	7.6	7.4	6.7	7.0	**7.3**

X-shaped Base and Back-support, and Double-beam on the Base						
Ease to Build	Weight Support	Materials	Comfort	Aesthetics	Dimensions	OVERALL AVERAGE
6.1	7.7	6.0	6.4	8.0	7.7	**7.0**

Hourglass-shaped Base with Folding Back Support						
Ease to Build	Weight Support	Materials	Comfort	Aesthetics	Dimensions	OVERALL AVERAGE
7.9	7.9	8.4	7.9	7.6	7.1	**7.8**

Final Design Prototypes

The highest-scoring design was made into a final design prototype. The initial 1:4 prototype (Figure 3) performed poorly in **back-support sturdiness**. After various design iterations, a more steady seat and back-support design was implemented on the a final 1:4 prototype (Figure 4).

Figure 3. Initial 1:4 Prototype

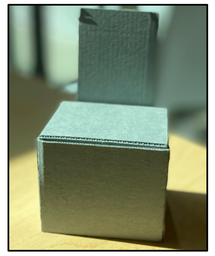
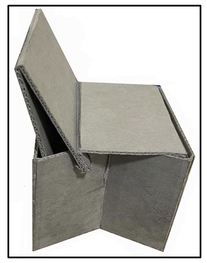


Figure 4. Final 1:4 Prototype



Final Design

The final layout on the cardboard (Figure 5) is created according to the final 1:4 prototype. This design (Figure 6) includes a base with **interlocked triangles** for ease of assembly and support. An **low, angled back rest** connected to the seat improves the stability of the structure while reducing the amount of cardboard used. These decisions make the chair **easy to assemble** with only 3 pieces, accommodating to **children of all sizes**, and **aesthetically pleasing**.

Figure 5. Final Cardboard Layout

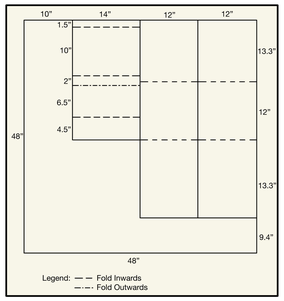


Figure 6. Final design

Figure 7 and Figure 8 demonstrate the dimensions of the chair once assembled.

Figure 7 focuses on the insertion of the seat and back-support, as well as the overall appearance and dimensions of the chair.

Figure 8 focuses on the assembly of the base, where two triangular structures intersect and allow the seat and back-support to be inserted.

Figure 7. Front View

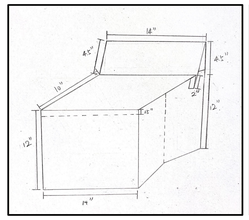
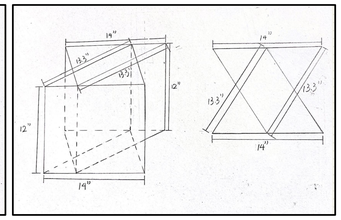


Figure 8. Front and Top View (base)



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